

Predicting Tennis Match Outcomes Using the Bradley–Terry Model

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Abstract

In this report we apply the Bradley–Terry model to predict the outcomes of ATP tennis matches using match data from the 2024 and 2025 seasons, consisting of 584 unique players and 5,976 matches. Each player is assigned a log-strength parameter, estimated by maximum a posteriori (MAP) estimation, which combines the Bradley–Terry likelihood with a Gaussian prior centred at zero. The prior is applied according to the number of matches each player has participated in, resolving the separation problem that arises with maximum likelihood estimation for players with extreme win–loss records, and shrinking players who contested few matches towards a neutral score of zero. The negative log-posterior is minimised efficiently using gradient-based optimisation in R, with the gradient supplied analytically. Model performance is assessed through k -fold cross-validation with $k = 10$, which produced an accuracy of 63.55% and a log loss of 0.627, reasonable results for a model based solely on match outcomes. Uncertainty in the estimated strengths is conveyed through standard errors derived from the Hessian matrix of the negative log posterior. The resulting strength estimates are similar to the ATP world rankings, with Jannik Sinner identified as the strongest player in the dataset.

Contents

1	Introduction	3
1.1	The Bradley–Terry Model	3
1.2	Application to Tennis	3
1.3	The Dataset	3
2	Model Description	3
2.1	Model Formulation	3
2.2	The Likelihood	4
2.3	Using MAP Estimation	4
2.4	Choosing a Suitable Prior	4
2.5	The Gradient	5
2.6	Optimisation	5
2.7	Implementation in R	6
3	Cross-Validation to Improve Accuracy and Log Loss	6
3.1	Accuracy and Log Loss	6
3.2	Cross-Validation on k-folds	7
3.3	Using Grid Search and Cross Validation to Choose λ	7

4	Results	7
4.1	Estimated Player Strengths	7
4.2	Comparing Our Results to the ATP Rankings	8
4.3	Accuracy and Log Loss Results	8
4.4	Uncertainty Estimates	9
5	Discussion and Limitations	11
5.1	Limitations of the Model	11
5.2	Future Improvements	11
6	Conclusion	11

1 Introduction

1.1 The Bradley–Terry Model

The Bradley-Terry model is a probability model for the outcome of pairwise comparisons between items, teams, or objects [1]. Given a pair of items i and j taken from some population, it estimates the probability that i beats j as:

$$\Pr(i > j) = \frac{p_i}{p_i + p_j} \quad (1)$$

where $p_i > 0$ is a positive real-valued score assigned to player i . The comparison $i > j$ can be read as “ i is preferred to j ”, “ i ranks higher than j ”, or “ i beats j ”, depending on the application [2] & [3].

The model can also be parametrised using exponential score functions $p_i = e^{\beta_i}$, giving: [4]

$$\Pr(i > j) = \frac{e^{\beta_i}}{e^{\beta_i} + e^{\beta_j}} = \frac{1}{1 + e^{\beta_j - \beta_i}} \quad (2)$$

1.2 Application to Tennis

Tennis results can be predicted effectively using a Bradley Terry model as many players play a high volume of games each season, and the only potential outcomes for each player are win or loss. In addition, the model naturally reflects the significance of each result: an upset, in which a lower-rated player defeats a higher-rated one, produces a stronger adjustment to both players’ strengths than an expected win between closely matched opponents, allowing the model to respond appropriately to informative results.

1.3 The Dataset

For our project we used ATP data from the 2024 and 2025 seasons. The data consisted of 584 unique players and represented 5976 matches, sorted into a CSV file [5]. This incorporated a large number of games for elite players (often over 100), although also caused a problem, as many players included in the data had played very few matches. We had to apply shrinkage on our Bradley Terry model in order to account for this.

2 Model Description

2.1 Model Formulation

We gave every player an initial ability score of zero. The log-strength score of player i is given by θ_i , for each player. Our code sorted through the data and pulled each unique matchup, counting the number of times each matchup between two specific players i and j occurred, and the number of wins for each player. This meant we had just two columns representing each unique matchup. For our project, it was more practical to use the exponential parametrisation of the formula:

$$\Pr(i > j) = \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}} = \frac{1}{1 + e^{\theta_j - \theta_i}} \quad (3)$$

The reason for this is that it allows for negative player scores, and also avoids the division by zero issue that the non-parameterised formula would give.

2.2 The Likelihood

Maximum Likelihood Estimation is a method of estimating the parameters of a statistical model by finding the values that make the observed data as probable as possible. We attempted to use this to compute the strength score of each player. As a result of the above, our likelihood is given by:

$$L(\theta_i, \theta_j) = \left(\frac{e^{\theta_i}}{e^{\theta_j} + e^{\theta_i}}\right)^{w_i} \left(\frac{e^{\theta_j}}{e^{\theta_i} + e^{\theta_j}}\right)^{w_j}$$

where θ_i, θ_j are the abilities of each player in the matchup, and w_i is the number of times player i beats player j, and w_j is the number of times player j beats player i.

We needed to use the `optim` function in R later on, which minimises but does not maximise. Maximising the log likelihood and minimising the negative log likelihood are equivalent approaches. Therefore, we needed the negative log-likelihood, which will be minimised. Thus, taking the log, and negating it, we got the following negative log likelihood:

$$\ln L(\theta_i, \theta_j) = w_i \left(\log(e^{\theta_i} + e^{\theta_j}) - \theta_i \right) + w_j \left(\log(e^{\theta_i} + e^{\theta_j}) - \theta_j \right)$$

2.3 Using MAP Estimation

Using the approach of Maximum Likelihood Estimation would not accurately account for players who played a low number of games and either won all of them or lost all of them, as these players would be given extremely high or low scores which would not be representative of their status in the world rankings. To overcome this issue we used a Maximum a Posteriori (MAP) Estimate. The Maximum a Posteriori (MAP) estimate is the value of θ that maximises the posterior distribution:

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} f(\theta | X_1, X_2, \dots, X_n)$$

By Bayes' theorem, the posterior is proportional to the likelihood multiplied by the prior:

$$\begin{array}{ccc} \text{Posterior} & & \text{Likelihood} \quad \text{Prior} \\ \text{(after data)} & & \text{(data given } \theta \text{)} \quad \text{(before data)} \\ \downarrow & & \downarrow \quad \downarrow \\ P(\theta | \text{data}) & = & \frac{P(\text{data} | \theta) P(\theta)}{P(\text{data})} \end{array}$$

Maximising this posterior balances fitting the observed match data (the likelihood) against the belief that strengths stay near zero (the prior), which is what resolves the extreme scores described above [6].

2.4 Choosing a Suitable Prior

We needed to choose a prior which would allow the likelihood to dominate for players who had played a large number of games, while shrinking the scores of players who had played a small number of games towards zero. We chose to use a Gaussian prior centered at zero, given by:

$$\frac{1}{2} \sum_i (\theta_i / \sigma_i)^2$$

This reflects our prior knowledge that, while some players are elite, most have strengths near the average. Combining this prior with the likelihood via Bayes' theorem yields the MAP estimate, where the prior term penalises extreme strength values and shrinks each estimate towards zero.

In order to avoid dividing by extreme values, we let $\lambda_i = 1/\sigma_i$. Thus, its influence on a given player's estimate was inversely proportional to the number of matches they had played. Scores

of players who played a large number of matches were still be dominated by the likelihood. However, scores of players who played fewer matches were pulled towards zero. In our model we used two separate values of λ - one value for players who had played more than five matches and a larger value for players who played 5 matches or less as this seemed to account for all obvious outliers. The prior term is added to the negative log-likelihood to create a new term called the 'negative log posterior', given by:

$$\text{NLP}(\boldsymbol{\theta}) = \sum_{(i,j)} \left[w_i \left(\log(e^{\theta_i} + e^{\theta_j}) - \theta_i \right) + w_j \left(\log(e^{\theta_i} + e^{\theta_j}) - \theta_j \right) \right] + \frac{1}{2} \sum_i (\theta_i \lambda_i)^2$$

Players who play a large number of games will have a correspondingly large likelihood term, while the prior will have greater effect on players who participated in fewer matches.

2.5 The Gradient

To minimise the negative log-likelihood efficiently, we supplied the gradient analytically. Differentiating the each term with respect to θ_i requires the derivative of the shared denominator, $\log(e^{\theta_i} + e^{\theta_j})$, which is itself the win probability

$$\frac{\partial}{\partial \theta_i} \log(e^{\theta_i} + e^{\theta_j}) = \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}} = P(i \text{ beats } j).$$

Thus, the derivative of the negative log-likelihood with respect to θ_i for a single matchup between players i and j is

$$\frac{\partial}{\partial \theta_i} = n_{ij} P(i \text{ beats } j) - w_i,$$

where $n_{ij} = w_i + w_j$ is the total number of matches played between the two players. The gradient is therefore the difference between expected and observed wins. When a player has won more matches than their current strength predicts, this difference is negative and the optimiser increases their strength; when they have won fewer, it decreases it. When we add the prior term to the above gradient, we get the 'negative log-posterior' gradient, given by:

$$\frac{\partial \text{NLP}}{\partial \theta_i} = \sum_{j:(i,j)} [n_{ij} P(i \text{ beats } j) - w_i] + \theta_i \lambda_i^2$$

Each player's full gradient is obtained by adding together these contributions across every matchup in which they appear.

2.6 Optimisation

Supplying this gradient in closed form allowed the BFGS optimiser function in R to converge rapidly. Setting the gradient to zero to find the optimal strengths gave a system of equations in which each player's strength appeared on both sides, with no closed-form solution for more than two players. Instead, we estimated the strengths numerically. The objective was minimised using R's `optim` function with the BFGS method, an algorithm similar to Newton's method that began from our initial guess of zero for every player and iteratively adjusted the full vector of strengths to reduce the negative log posterior at each step. At every iteration the gradient indicated the direction of steepest descent, and BFGS built an approximation to the curvature of the objective, allowing it to converge in relatively few iterations. The procedure ended when the gradient was arbitrarily close to zero, at which point the returned vector of strengths was given as the MAP estimate.

2.7 Implementation in R

The model was implemented in R as a sequence of modular functions. Before fitting, the match data was processed so that each player was mapped to a unique integer index rather than being referred to by name, as this made computation faster. Next, we constructed a function which aggregated the data into head-to-head summaries, where for every pair of players who met, we recorded the two integer indices together with the number of times each player won. This reduced the data from one row per match to one row per unique matchup, which allowed the objective to be evaluated compactly.

Each player was assigned a regularisation parameter according to the number of matches they have played, as described above. The negative log posterior was minimised using the `optim` function with the BFGS method, supplied with both the objective and the gradient, and returning the fitted player strengths. The negative log-likelihood and its gradient were written in vectorised form for efficiency.

3 Cross-Validation to Improve Accuracy and Log Loss

3.1 Accuracy and Log Loss

To evaluate the model's predictive performance we used accuracy and log-loss.

Accuracy is a measure of the proportion of results which were predicted correctly by our model. For each match, the model predicts the player with the higher estimated strength as the winner, and this prediction is compared against the actual result. Accuracy is then the fraction of matches for which the prediction was correct, and is given by:

$$\text{Accuracy} = \frac{1}{N} \sum_{m=1}^N \mathbf{1}[\hat{y}_m = y_m]$$

where N is the number of matches, y_m is the true winner, $\hat{y}_m$ is the predicted winner and $\mathbf{1}$ is the indicator function which takes a value of 1 if the predicted winner and the true winner are the same, and zero otherwise.

Log Loss (also known as Binary Cross Entropy Loss) is the negative mean log of the probabilities p_m (the probabilities that each outcome actually occurred), given by:

$$\text{Log loss} = -\frac{1}{N} \sum_{m=1}^N \left[y_m \log p_m + (1 - y_m) \log(1 - p_m) \right]$$

where $y_m \in \{0, 1\}$ is the actual outcome and p_m is the predicted probability of the positive outcome. In our model, the winner is always stored in the first column, so the formula simplifies to:

$$\text{Log loss} = -\frac{1}{N} \sum_{m=1}^N \log p_m$$

We wanted to increase accuracy and decrease log loss as much as possible in our model, while still getting realistic results. We decided log loss was more telling than accuracy because it evaluates the quality of the model's predicted probabilities rather than just whether the favourite was picked correctly — rewarding confident correct predictions and penalising confident wrong predictions. We thought this better reflected what the Bradley-Terry model was actually producing.

3.2 Cross-Validation on k-folds

Cross-validation is a technique for estimating how well a model will perform on unseen data, by repeatedly training and testing it on different sub-sections of the available data [7]. For our model, we realised that when using just a single fixed training/test split on the data, the accuracy and log-loss figures was dependent heavily on which matches happen to fall into the test set. We felt results with a lower log-loss could be obtained by splitting the data into a certain number of equally sized folds and applying cross-validation. The match data was split into k equal folds, and for each fold the model was trained on the remaining $k - 1$ training folds, which were used to predict the test fold. For every test match, the player with the higher estimated strength was taken as the predicted winner and compared against the actual result. Each fold was used once as the test set, and the log loss was then averaged across all folds to give a single, more reliable estimate of the performance of the model. Performing k-fold cross validation helped to reduce log loss as this allowed for training on the entire data set and evaluated the model multiple times.

3.3 Using Grid Search and Cross Validation to Choose λ

To choose a suitable pair of values for $\lambda = (\lambda_{strong}, \lambda_{weak})$ for our prior term in the negative log posterior, we performed a grid search. We decided to use two separate λ values a larger/stronger value for players who played five games or less, and a smaller/weaker value for players who played more than five games. We defined a set of candidate values for λ for both groups of players and for each potential pair of values, evaluated the model's performance using 10-fold cross-validation, recording the cross-validated log loss. The log loss corresponding to this λ was stored in a matrix for the grid search, and the process was repeated for each subsequent λ . Once this process was complete, we would find the minimum log loss over the λ 's.

The issue with the above however was that for our $\lambda = (\lambda_{strong}, \lambda_{weak})$, we observed a pattern that as λ_{strong} got bigger and bigger we got smaller and smaller log loss values. These log loss values were decreasing at a very slow rate. With extremely harsh shrinkage we were losing information on the players because the stronger the shrinkage the more likely a player's ability score would be dragged to exactly zero.

In order to optimise for log loss without sacrificing too much information we set a threshold that other log loss values could fall into around the minimum log loss value. Any log loss value that fell within 0.1% of the minimum log loss was considered as a candidate for the log loss value of our optimal λ . Once this set of candidate pairs were found we chose the lambda pair which minimised the λ_{strong} parameter but also kept the λ_{weak} parameter at the same value as it was for the minimum log loss. The $\lambda = (\lambda_{strong}, \lambda_{weak})$ we landed on after this process was our optimal lambda pair. Our choice is $\lambda = (\lambda_{strong}, \lambda_{weak}) = (2.7, 0.96)$

4 Results

4.1 Estimated Player Strengths

Our model gave strength estimates for all 584 players in the data set. Jannik Sinner was the strongest player, according to the model, with a score of 3.655, averaged across all cross-validated folds, while Chun Hsin Tseng was chosen by the model as the weakest player, with a score of -0.3846. There was also a clear correlation between the number of matches each player played and their estimated strength. This makes sense as, in most cases, better players would progress further in tournaments, and hence play more matches than weaker players. The following image shows the distribution of scores for players who played at least five matches. It indicates that the majority of players have an estimated Bradley Terry strength between 0 and 1, which is in

line with our expectations

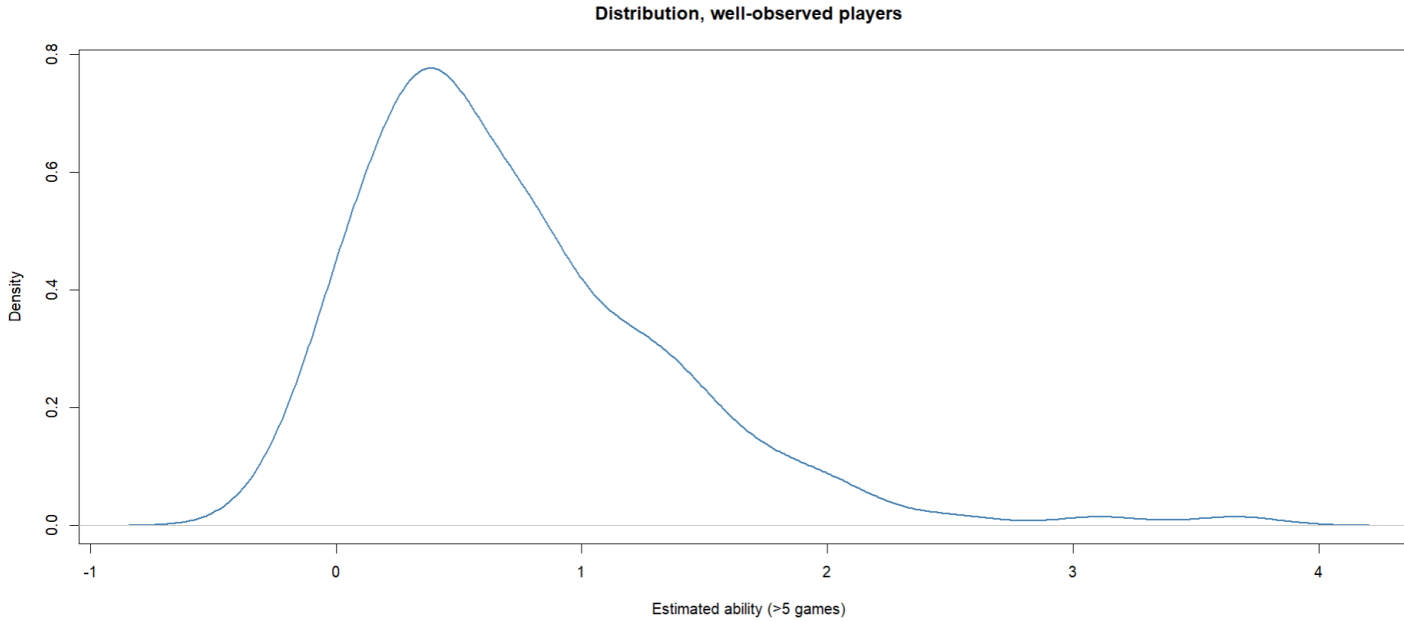


Figure 1: Distribution of scores for players who have played more than 5 games

4.2 Comparing Our Results to the ATP Rankings

The ATP ranks players by summing the points of each player from their best nineteen tournament results over the most recent 52 week period. A higher amount of points are rewarded for winning matches in Grand Slam competitions and the ATP finals than some of the less prestigious ATP competitions. The rankings are updated on a weekly basis so are subject to change regularly. This is a tournament importance based ranking system which adds extra weight to victories in more prestigious matches, but only accounts for one year worth of data. On the other hand, our model ranks players based entirely on matchup results, regardless of tournament, and account for two years worth of data. Despite this different scoring system, the ranking of players by the Bradley Terry model was similar to the ATP rankings from 5/1/2026. There were also some slight differences, such as Carlos Alcaraz being ranked the best player by the ATP, but the second best by our model. The top ten players in both scoring systems are printed below, with players who appear in the top ten of both models marked with a tick.

4.3 Accuracy and Log Loss Results

Using our cross-validated approach, we reduced the log-loss of the model to 0.627. The baseline score, if we treated every player as having an equal chance of winning in every unique matchup, would be $-\log(0.5) = 0.693$. Thus our model has a reasonably low log-loss considering it is based entirely upon match outcomes.

The predictive accuracy of our model was 62.76%. The predictive accuracy doesn't capture the full strength of this model alone since this value is still reasonable considering the model does not include any extra covariates.

We also used the ATP Rankings as of 5/1/2026 to predict match outcomes. For each matchup,

Bradley–Terry model			ATP Rankings 5/1/2026		
#	Player	Score	#	Player	Points
1	Jannik Sinner ✓	3.557	1	Carlos Alcaraz ✓	12,050
2	Carlos Alcaraz ✓	3.013	2	Jannik Sinner ✓	11,500
3	Novak Djokovic ✓	2.431	3	Alexander Zverev ✓	5,105
4	Alexander Zverev ✓	2.119	4	Novak Djokovic ✓	4,780
5	Alex de Minaur ✓	1.962	5	Felix Auger-Aliassime	4,190
6	Taylor Fritz ✓	1.957	6	Alex de Minaur ✓	4,080
7	Jack Draper ✓	1.886	7	Lorenzo Musetti	3,990
8	Tommy Paul	1.751	8	Ben Shelton	3,960
9	Daniil Medvedev	1.748	9	Taylor Fritz ✓	3,840
10	Grigor Dimitrov	1.730	10	Jack Draper ✓	2,990

we always selected the player with the highest rank as our prediction to win. This approach predicted the correct winner 63.52% of the time. Thus, our model has only a slightly worse level of accuracy to the approach of picking the higher ranked player every time. Despite comparing the our model’s predictive accuracy to the predictive accuracy of the ATP rankings we still believe log loss is a better metric for measuring model performance before measuring predictive power on unseen data.

To summarise, our humble Bradley-Terry model can predict with an accuracy within 1% of the official experts’ ATP rankings!

4.4 Uncertainty Estimates

The slight differences in the ranking of players between our model and the ATP system highlighted the variation at play when determining a player’s ability. The fitted strengths that the model gave were point estimates, but players who played less matches would have greater uncertainty around their scores, while a player observed over many matches is estimated far more precisely. To account for this, we computed a standard error for each player’s estimated strength. These were obtained from the Hessian matrix of the negative log posterior which the `optim` function returns at the optimum. The Hessian is the matrix of the second partial derivatives with respect to the player strengths.

The inverse of the Hessian approximates the covariance matrix of the estimated strengths. The standard error of each player’s strength was taken as the square root of the corresponding diagonal entry of the inverse of the Hessian:

$$SE(\hat{\theta}_i) = \sqrt{[H^{-1}]_{ii}},$$

where H is the Hessian of the negative log posterior evaluated at the fitted strengths. As expected, the standard errors of players who played five matches or less were quite small due to the stronger shrinkage. For players who played more than five games, an increase in the number of games played was correlated with a reduced standard error, as indicated by the following graph:

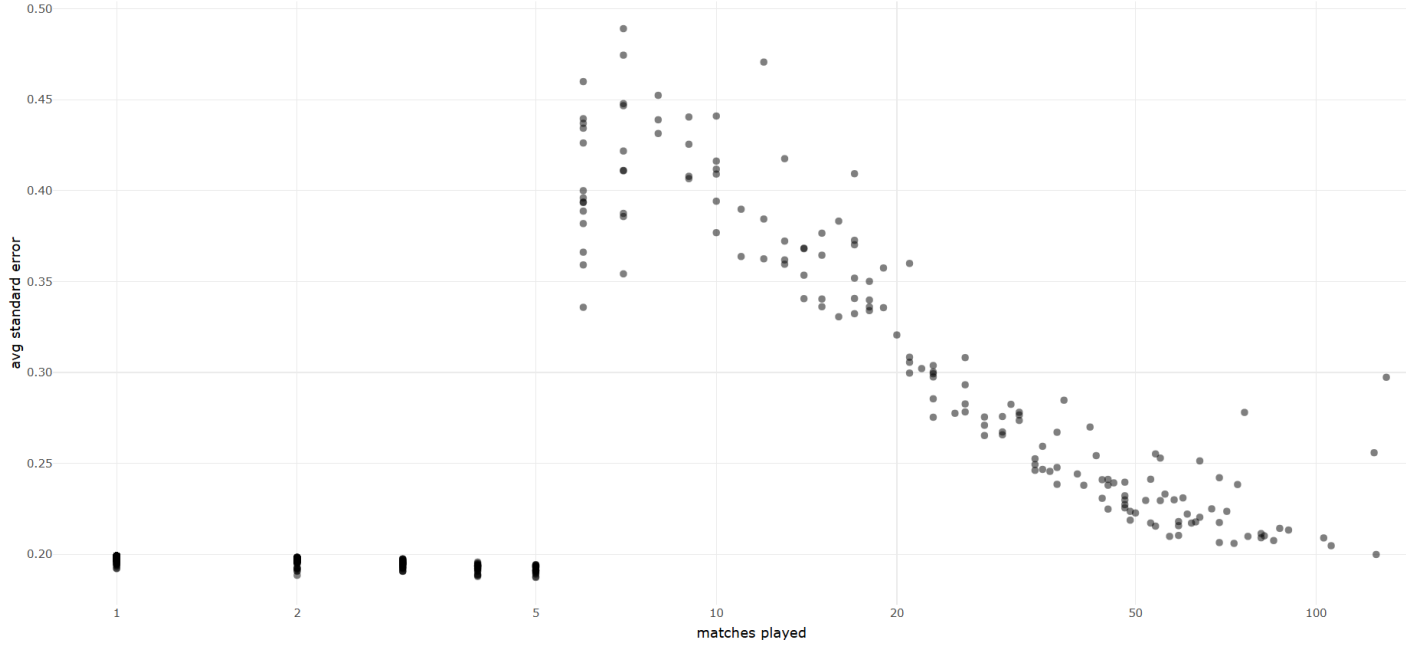


Figure 2: Average standard errors of players compared to their number of matches played, according to the model

This reflects the fact that the true strength scores of players who fall into the category of weak shrinkage but still played barely more than 5 games would vary more than those who appeared more frequently in the data. The following centipede plot shows the standard errors of the top 20 players' scores, estimated at the 95% confidence level.

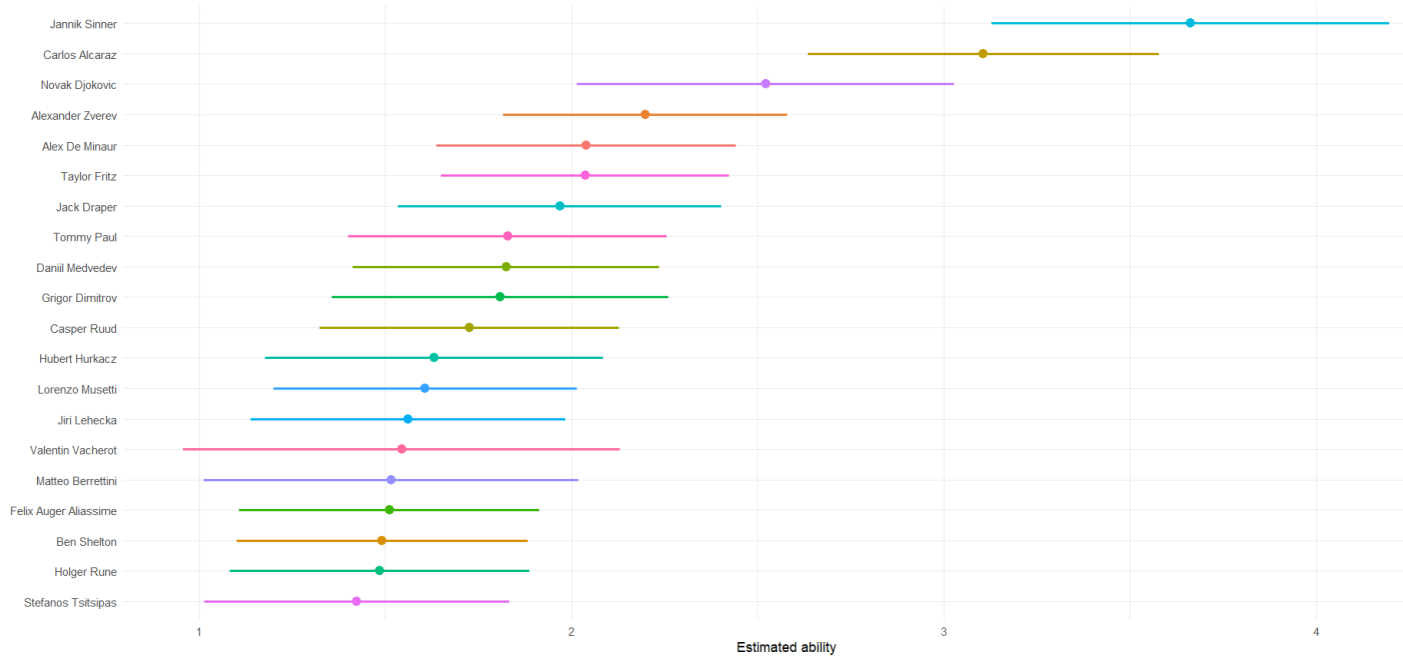


Figure 3: Scores and standard errors of top 20 players, according to the model

5 Discussion and Limitations

5.1 Limitations of the Model

While the model ranked players similarly to the ATP rankings, it did have some drawbacks and limitations. Firstly, our Bradley Terry Model was based entirely on previous match outcomes. In tennis, factors such as surface, injuries and player motivation play a large role in determining results, but we did not account for any of these extra covariates when building our model.

The use of a Gaussian prior in MAP estimation ensured outliers in the data were accounted for. However, for players who had played a low number of games, the model did not provide an obvious way to check if the scores provided were accurate. For example, for a player who played three matches and won all of them, our prior shrinks their ability heavily towards zero. The model does not provide us any way of knowing whether this score actually represents the player’s ability realistically, or if he is in fact an elite player who happened to play a low number of games.

The model also only allows for one strength score per player. If players were given multiple strength scores, perhaps time-weighted or one for each different surface, more accurate predictions could be achieved.

5.2 Future Improvements

If given more time to build the model, we could have included an extra surface covariate. We believe this would have significantly improved the accuracy, as many players tend to have a preferred surface where they perform better. This would have involved much more work as each player would have been given a unique ‘surface’ covariate value, for each different surface type.

In our model, match data is treated as equally important regardless of recency. A match played in January of 2024 is treated as equally important to a match played in December 2025. If we had more time, we could have time-weighted the data, giving more importance to the matches which were played in closer proximity to the test matches, for each of our k -folds. This would have once again been difficult and computationally expensive as each match would have to be given a different time-weight covariate.

6 Conclusion

In this report we applied the Bradley–Terry model to the problem of predicting the outcomes of ATP tennis matches. Using the exponential parameterisation of the formula, we estimated each player’s ability by maximum a posteriori estimation, combining the Bradley-Terry likelihood with a Gaussian prior. The prior, applied adaptively according to the number of matches each player had contested, resolved the problem that was occurring with players who had played a small number of games and produced sensible strength estimates for all players. The negative log posterior was minimised efficiently using gradient-based optimisation with an analytically derived gradient. The model’s performance was assessed through k -fold cross-validation.

The resulting model produced strength estimates that aligned well with our expectations, ranking the leading players in an order similar to the ATP ranking system. Using MAP Estimation and cross-validation allowed us to create a model which was close to what we believe to be the realistic ceiling for a model based solely on match outcomes. While the model could be improved in a number of ways described above, it demonstrates that a simple, well-regularised approach can predict tennis outcomes with reasonable accuracy and log-loss levels.

References

- [1] R. A. Bradley and M. E. Terry, *Rank analysis of incomplete block designs: I. The method of paired comparisons*, *Biometrika*, vol. 39, no. 3/4, pp. 324–345, 1952.
- [2] M. E. Glickman, *Parameter estimation in large dynamic paired comparison experiments*, *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, vol. 48, no. 3, pp. 377–394, 1999.
- [3] F. Santi and N. Friel, *Stochastic variational inference for dynamic Bradley–Terry models*, arXiv preprint arXiv:2511.03467, 2025.
- [4] M. Cattelan, C. Varin, and D. Firth, *Dynamic Bradley–Terry modelling of sports tournaments*, *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, vol. 62, no. 1, pp. 135–150, 2013.
- [5] J. Sackmann, *tennis_atp: ATP Tennis Rankings, Results, and Stats*, GitHub repository, 2024. Available: https://github.com/JeffSackmann/tennis_atp
- [6] L. Yan, C. Piech, M. Sahami, and J. Cain, *Maximum a Posteriori (MAP) Estimation*, CS109: Probability for Computer Scientists, Lecture 22, Stanford University, Spring 2021.
- [7] B. Efron and T. Hastie, *Computer Age Statistical Inference: Algorithms, Evidence, and Data Science*, Cambridge University Press, 2016.